Perfect-Power Obstructions and Sparse Powerful Values in Cyclic Decimal Concatenations

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Abstract

We prove local obstructions and a quantitative density-zero bound for powerful values among cyclic concatenations of initial positive integers. For decimal digit lengths with $v_3(d) = 1$, a cubic reduction gives an exact local classification: 131 classes of m modulo 243 exclude all rotations over $\mathbb{Q}_3$; every other class has a local square, cube, or fifth power among the first four cuts. At complete digit boundaries, exact valuations and characters exclude a set of digit lengths of lower density exceeding 95.5995%. A general lifting theorem gives local power equivalences and a quadratic reduction on entire digit intervals. Along suitable progressions of digit lengths, the proportion of m for which no rotation is powerful can be arbitrarily close to one. The quantitative sparsity bound holds in every fixed integer base.

1 Statements

In base $g \geq 2$, let W_m concatenate $1, 2, \ldots, m$, and let $D(m)$ be its digit length, with $W_0 = D(0) = 0$. For $0 \leq k < m$, moving the first k whole integer blocks to the end gives

$$R_{m,k} = g^{D(k)} W_m - (g^{D(m)} - 1) W_k. \tag{1}$$

In base ten these numbers, in increasing order, form OEIS A001292 [1]. Kashihara conjectured that none except 1 is a perfect power. Ripà [2] excludes $m \equiv 2, 3, 5, 6 \pmod 9$ by a digit-sum argument. The full conjecture remains open. We give a uniform sparsity bound and local obstructions for complete digit boundaries.

Recall that a positive integer is *powerful* if every prime divisor has exponent at least two. Every nontrivial perfect power is powerful. Logarithms in the following theorem and in Section 5 are natural.

Theorem 1 (Quantitative sparsity)**.** *For every fixed integer base $g \geq 2$,*

$$\#\{(m,k) : 2 \leq m \leq X,\ 0 \leq k < m,\ R_{m,k} \text{ powerful}\} \ll_g \frac{X^2 (\log\log X)^2}{\log X}. \tag{2}$$

For A001292, the corresponding count among the first N terms is $O(N(\log\log N)^2/\log N)$.

For the next result the base is ten and v_3 denotes the 3-adic valuation.

Theorem 2 (Complete digit boundaries)**.** *Let $d \geq 1$, $m = 10^d - 1$, $h = v_3(d)$, and $u = d/3^h$. Every rotation satisfies*

$$v_3(R_{m,k}) = h + 2, \qquad \frac{R_{m,k}}{3^{h+2}} \equiv u(2 + 6d) \pmod 9. \tag{3}$$

Corollary 1. *Put $v = h + 2$ and let $\mathcal{P}(v)$ be its set of prime divisors. No rotation at $m = 10^d - 1$ is a perfect power in any of the following cases:*

$\mathcal{P}(v)$	*condition on* u
$\{2\}$	$u \equiv 1 \pmod 3$
$\{3\}$	$u \equiv 1, 2, 7, 8 \pmod 9$
$\{2, 3\}$	$u \equiv 1, 7 \pmod 9$.

This includes $d \equiv 3, 6, 9, 21, 24 \pmod{27}$ and $d = 81$.

Proof. A prime exponent of a perfect power must divide v. A square requires v even and its normalized unit congruent to 1 modulo 3; by (3) this requires $u \equiv 2 \pmod 3$. A cube requires $3 \mid v$, hence $h \geq 1$ and $3 \mid d$. Its normalized unit must be $\pm 1 \pmod 9$, so $2u \equiv \pm 1 \pmod 9$, or $u \equiv 4, 5 \pmod 9$. The table excludes every available prime exponent. □

For the additional boundary primes, put

$$(p, s) \in \{(37, 3), (333667, 9)\}, \qquad C_s = \frac{9(11 - 10s)}{200} \pmod p.$$

Define

$$A_0 = 0, \qquad A_r = -9 \sum_{j=1}^{r} \frac{10^{j-1}}{10^j - 1} \pmod p \quad (0 < r < s).$$

All denominators here are units modulo p; $C_3 = 33$ and $C_9 = 221885$.

Theorem 3 (Two affine residue formulas)**.** *For every $d \geq 1$, write $d = sn + r$, $0 \leq r < s$. Then*

$$V_d := W_{10^d - 1} \equiv C_s n + A_r \pmod p, \qquad R_{10^d - 1, k} \equiv 10^{D(k)} V_d \pmod p. \tag{4}$$

Theorem 4 (General boundary-prime lifting)**.** *Let $p \geq 7$ be prime, $r = \operatorname{ord}_p(10)$, and choose P with*

$$r \mid P, \qquad 10^P \equiv 1 \pmod{9r}, \qquad p \nmid P, \qquad C := V_P \not\equiv 0 \pmod p.$$

For every positive d divisible by P, set $a = v_p(d)$. Every whole-block rotation satisfies

$$v_p(R_{10^d - 1, k}) = a, \qquad \frac{R_{10^d - 1, k}}{p^a} \equiv 10^{D(k)} C \frac{d}{P p^a} \pmod p. \tag{5}$$

In particular, $d = Ppt$ with $p \nmid t$ excludes all perfect powers. More generally, every possible prime exponent divides both $v_3(d) + 2$ and $v_p(d)$, for each applicable pair (p, P). The six primes in the table of Section 4 all satisfy the hypotheses. For the two affine formulas take $P = s$ and $C = C_s$. If a prime exponent ℓ divides $(p - 1)/r$, then $Cd/(Pp^a)$ must also be an ℓth power modulo p. In particular, $d = 111t$ with $37 \nmid t$ and $d = 3003003t$ with $333667 \nmid t$ are excluded. The new choice $p = 757, P = 27$ also excludes $d = 327024 = 27 \cdot 757 \cdot 16$, since the valuations of every rotation at 3 and 757 are respectively 5 and 1.

Theorem 5 (Reduction on an entire digit interval)**.** *Let $p \geq 7$ be prime, $r = \operatorname{ord}_p(10)$, and $r \mid P$, $10^P \equiv 1 \pmod{9r}$, $p \nmid P$. Put $C = V_P \pmod p$, allowing $C = 0$. For $P \mid d$, $a = v_p(d)$, every $10^{d-1} \leq m < 10^d$ and $0 \leq k < m$ satisfy*

$$R_{m,k} \equiv 10^{D(k)} \left(\frac{m(m+1)}{2} + C \frac{d}{P} \right) \pmod{p^{a+1}}. \tag{6}$$

If $Pp \mid d$, exactly $2(p - 1)$ residue classes $m \pmod{p^2}$ give $v_p(R_{m,k}) = 1$ for every cut k.

For example, if $d = 111u$, all rotations have valuation one at 37 whenever

$$m \equiv 0, -1 \pmod{37}, \qquad m \not\equiv 296u,\ 1073u - 1 \pmod{1369}.$$

These are 72 residue classes modulo 1369 in every such digit interval.

Corollary 2 (Arbitrarily high coverage of whole rotation blocks)**.** *For every $\varepsilon > 0$ there are positive integers L, d_0 such that, for every $d \geq d_0$ divisible by L, at least $(1 - \varepsilon)9 \cdot 10^{d-1}$ choices of d-digit m have no powerful integer among their rotations.*

Corollary 3 (Density of excluded complete boundaries)**.** *The set of positive integers d for which every rotation at $m = 10^d - 1$ is not a nontrivial perfect power has lower natural density at least*

$$\rho_* = 0.9559952715065166\ldots > 0.95599527150.$$

Theorem 6 (Further complete-boundary exclusions)**.** *No rotation at $m = 10^d - 1$ is a nontrivial perfect power when $1 \leq d \leq 27$, or when $d = 54$ or 243. The same conclusion holds for*

$$\begin{aligned} &d \equiv 12, 15 \pmod{333}, \\ &d \equiv 27 \pmod{37249072371}, \qquad d \equiv 243 \pmod{25475666076}. \end{aligned} \tag{7}$$

The finite inputs to this theorem are exact modular certificates; Section 4 gives their verification and the unbounded period argument.

2 Exact 3-adic values at complete boundaries

Write $V_d = W_{10^d-1}$ and $S(d) = D(10^d - 1)$. We will prove the stronger intermediate congruence

$$\frac{V_d}{9d} \in \mathbb{Z}_3, \qquad \frac{V_d}{9d} \equiv 2 + 6d \pmod 9. \tag{8}$$

Quotients by d in this section are taken in $\mathbb{Q}_3$.

Lemma 1 (Averaging power series)**.** *For integers $d \geq 1$ and $n \geq 1$,*

$$v_3\left(\frac{1}{d}\sum_{j=1}^{d} j^n\right) \geq -\lfloor \log_3(n+1) \rfloor. \tag{9}$$

Suppose $f(X) = \sum_{n\geq 0} c_n X^n$ has $v_3(c_n) \geq n$, $c_0 \equiv 5 \pmod 9$, $c_1 \equiv 3 \pmod 9$, and $v_3(c_2) \geq 3$. Then

$$\frac{1}{d}\sum_{j=1}^{d} f(j) \equiv 2 + 6d \pmod 9. \tag{10}$$

Proof. Using Stirling numbers of the second kind gives the exact identity

$$\frac{1}{d}\sum_{j=1}^{d} j^n = \sum_{r=0}^{n} \left\{ {n+1 \atop r+1} \right\} \frac{r!}{r+1}\binom{d-1}{r}.$$

Each denominator loses at most $\lfloor \log_3(n+1) \rfloor$ factors of 3, proving (9). For $n \geq 3$, $n - \lfloor \log_3(n+1) \rfloor \geq 2$, so all these terms vanish modulo 9 after averaging. The same holds for $n = 2$ by its stronger coefficient bound. Only $c_0 + c_1(d+1)/2$ remains, congruent to $5 + 3(d+1)/2 = 2 + 6d \pmod 9$. The coefficient bounds also justify convergence and termwise averaging. □

Define the 3-adic logarithm explicitly by

$$\ell = \sum_{r\geq 1} \frac{(-1)^{r+1} 9^r}{r}, \qquad v_3(\ell) = 2.$$

In the following power series put

$$E = e^{\ell X}, \quad a = E/10, \quad q = 9a, \quad b = E - 1, \quad S(X) = \frac{(9X-1)E + 1}{9},$$

and define

$$\frac{U(X)}{9} = a^2 + \frac{ab+1}{9}\sum_{r\geq 0}\binom{q}{r+2} b^r, \qquad f(X) = \frac{U(X)}{9} e^{-\ell S(X)}. \tag{11}$$

At every positive integer j, $S(j)$ is the digit length of V_j, and $U(j)$ concatenates all j-digit integers. Indeed, with $a = 10^{j-1}$, $b = 10^j - 1$, and $q = 9a$, the usual arithmetic-geometric sum gives

$$b^2 U(j) = (ab+1)(1+b)^q - ((1+b)b + 1).$$

Expanding $(1+b)^q$ cancels its constant and linear terms and gives (11), since $a + q = 10^j$ and $aq = 9a^2$.

Lemma 2. *The coefficients of f satisfy all hypotheses of Lemma 1.*

Proof. Let $\mathcal{A}$ consist of series whose degree-n coefficient has valuation at least n. It is closed under addition and multiplication. In the rth summand of $U/9$, the factor b^r makes all degrees below r zero. The factor q in $\binom{q}{r+2}$ cancels the displayed denominator 9. The remaining expression is a polynomial in E with 3-adic denominators bounded by $(r+2)!$. Its degree-n coefficient, for $r \leq n$, therefore has valuation at least

$$2n - v_3(n!) - v_3((r+2)!) \geq n.$$

Here $v_3(n!) \leq (n-1)/2$ for $n \geq 1$, and $v_3((r+2)!) \leq (n+1)/2$; the degree-zero case is immediate. The term a^2 satisfies the same bound. Thus $U/9 \in \mathcal{A}$.

The constant term of S is zero. Its linear coefficient is $1-\ell/9$, of valuation at least 2, and for $n \geq 2$ its coefficient is $\ell^{n-1}(n-\ell/9)/n!$, of valuation at least n. Hence $S \in \mathcal{A}$. Expanding $e^{-\ell S}$ preserves these bounds: its rth exponential term gains $2r - v_3(r!) \geq 0$ factors of 3. Consequently $f \in \mathcal{A}$.

Expansion through degree two in (11) gives

$$\begin{aligned} c_0 &= \frac{1}{200}, \\ c_1 &= \frac{169\ell}{3000} + \frac{\ell^2}{1800}, \\ c_2 &= \frac{671\ell^2}{80000} + \frac{353\ell^3}{54000} + \frac{\ell^4}{32400}. \end{aligned}$$

Since $\ell/9 \equiv 1 \pmod 9$, these imply $c_0 \equiv 5 \pmod 9$, $c_1 \equiv 3 \pmod 9$, and $v_3(c_2) \geq 3$. □

Proof of Theorem 2. The concatenation of the complete digit blocks yields

$$V_d = 10^{S(d)} \sum_{j=1}^{d} U(j) 10^{-S(j)} = 9\, 10^{S(d)} \sum_{j=1}^{d} f(j).$$

Lemma 1 and $10^{S(d)} \equiv 1 \pmod 9$ prove (8). Its right side is a 3-adic unit, so $v_3(V_d) = h+2$.

To transfer the normalized unit to every rotation, note that

$$S(d) = \frac{9d(d+1)}{2} \left(1 + \sum_{r=1}^{d-1} \frac{2 \cdot 9^r}{r+2} \binom{d-1}{r} \right). \tag{12}$$

This follows by expanding $S(d) = 9 \sum_{j=1}^{d} j 10^{j-1}$ in powers of 9. Each term after 1 in the bracket is divisible by 3; hence $v_3(S(d)) = 2 + h + v_3(d+1)$. Lifting the exponent gives

$$v_3(10^{S(d)} - 1) = 4 + h + v_3(d+1).$$

In (1), $10^{D(k)} \equiv 1 \pmod 9$ and the second term is divisible by 3^{h+4}. Thus $R_{m,k} \equiv V_d \pmod{3^{h+4}}$. Combining this with (8) proves both assertions in (3). □

A cubic reduction on digit intervals

Write $T_x = x(x+1)/2$.

Theorem 7 (Uniform cubic congruence)**.** *Let m have d decimal digits and $h = v_3(d) \geq 1$. For every whole-block cut,*

$$10^{-D(k)} R_{m,k} \equiv T_m + \frac{3d}{2}(m^3 - m + 9) - 9dmT_k \pmod{3^{h+4}}. \tag{13}$$

In particular, if $3^{h+2} \mid m$, all rotations satisfy

$$R_{m,k} \equiv \frac{m + 27d}{2} \pmod{3^{h+4}}. \tag{14}$$

Proof. Put $B = 10^d$, $\delta = B - 1$, and $t = B - 1 - m$. The exact tail identity in the proof of Theorem 5, expanded through first order in δ, gives

$$W_m \equiv V_d + \frac{t(t-1)}{2} - \delta \frac{t(t^2+5)}{6} \pmod{3^{2h+4}}.$$

Here the first derivative of the tail at $B = 1$ is $t(t+1)(5-2t)/6$; combining it with the factor B^{-t} gives the displayed integer coefficient. The omitted term δV_d also has valuation at least $2h+4$. Substitute $t = \delta - m$. Any loss from the denominator 6 leaves the discarded quadratic terms of valuation at least $2h+3 \geq h+4$. The binomial expansion gives $\delta \equiv 9d \pmod{3^{h+4}}$, while Theorem 2 gives $V_d \equiv 18d \pmod{3^{h+4}}$. Consequently

$$W_m \equiv T_m + \frac{3d}{2}(m^3 - m + 9) \pmod{3^{h+4}}.$$

Also $D(m) = d(m+1) - \delta/9 \equiv dm \pmod{3^{h+2}}$, so $10^{D(m)} - 1 \equiv 9dm \pmod{3^{h+4}}$. Finally $10^{-D(k)} W_k \equiv T_k \pmod 9$ by digit sums. The rotation identity proves (13). If $3^{h+2} \mid m$, the cubic, quadratic, and cut-dependent terms vanish at the stated precision except $m/2 + 27d/2$. Multiplication by $10^{D(k)} \equiv 1 \pmod 9$ preserves this residue, proving (14). □

Corollary 4 (Complete local classification when $v_3(d) = 1$)**.** *Let $v_3(d) = 1$ and $u = d/3$. All rotations of a d-digit m fail to be prime-exponent powers over $\mathbb{Q}_3$ exactly when at least one of the following holds:*

$$m \bmod 9 \in \{2,3,5,6\}, \qquad m \bmod 27 \in \{9,17\}, \qquad m \bmod 243 \in C_u,$$

where

$$C_u = \begin{cases} \{0,27,54,135,189\}, & u \equiv 1 \pmod 3, \\ \{54,108,162,189,216\}, & u \equiv 2 \pmod 3. \end{cases}$$

These are exactly 131 residue classes modulo 243, and every rotation in these classes is excluded from integer perfect powers. In every other class, some $k \in \{0,1,2,3\}$ and $e \in \{2,3,5\}$ satisfy $R_{m,k} \in (\mathbb{Q}_3^\times)^e$.

Proof. A 3-adic unit is a square exactly when it is 1 modulo 3, a cube exactly when it is ± 1 modulo 9, and always a fifth power. The square and fifth-power assertions follow from Hensel's lemma. For cubes, $(1+3x)^3 = 1+9(x+3x^2+3x^3)$, whose expression in parentheses has derivative 1 modulo 3, proves sufficiency; necessity is immediate. In each case the valuation must also be divisible by the exponent. The factor $10^{D(k)}$ preserves these three power criteria.

Modulo 27, (13) reduces to T_m. The first two listed sets have valuation one, or valuation two with nonsquare unit. Outside them, $m \equiv 1 \pmod 3$ gives a fifth-power unit, and $m \equiv 8, 18 \pmod{27}$ gives a square. It remains to consider $m \equiv 0, -1 \pmod{27}$.

For $m \equiv 0 \pmod{27}$, (14) becomes $R_{m,k}/27 \equiv (m/27 + 3u)/2 \pmod 9$. The nine classes of $m/27$ give precisely the five exclusions C_u; every other nonzero residue gives a square or cube. If this residue is zero, $m \equiv -81u \pmod{243}$. The binomial expansion gives $v_3((10^d-1)/(9d)-1) = 2$, so $v_3(D(m)) = 3$ and $v_3(10^{D(m)}-1) = 5$. Since $R_{m,1} = 10R_{m,0} - (10^{D(m)}-1)$, either $R_{m,0}$ or $R_{m,1}$ has valuation exactly 5 and is a fifth power in $\mathbb{Q}_3$.

For $m = 27z - 1$, (13) gives $R_{m,k}/27 \equiv (3u - z)/2 + uT_k \pmod 9$. At $k = 0, 2, 3$, these residues form an entire coset modulo 3 in $\mathbb{Z}/9\mathbb{Z}$. That coset contains 1, 3, or 8, giving a cube or a square. All these cuts exist because m has at least three digits. Finally the three exclusion sets are disjoint and contain $108 + 18 + 5 = 131$ classes modulo 243. □

More generally, (14) shows that if $h+3$ is a power of 2, $d/3^h \equiv 1 \pmod 3$, and $3^{h+4} \mid m$, then every rotation is excluded: its valuation is $h+3$ and its normalized unit is 2 modulo 3.

A finite sufficient test at every higher valuation

For $P \geq 2$, call a nonzero residue $a \pmod{3^P}$ *certifiably excluded* if, writing $b = v_3(a) < P$ and $w = a/3^b$, all the following hold:

1. $b > 0$, and every prime divisor of b belongs to $\{2,3\}$;
2. if $2 \mid b$, then $w \equiv 2 \pmod 3$;
3. if $3 \mid b$, then $P - b \geq 2$ and $w \not\equiv \pm 1 \pmod 9$.

The zero residue is never certifiably excluded. The conditions depend only on the residue, not its representative.

Proposition 1 (A sufficient nine-residue test)**.** *Let m have d decimal digits, $h = v_3(d) \geq 2$, and $P = h + 4$. Set*

$$A_j = T_m + \frac{3d}{2}(m^3 - m + 9) - 9dmT_j \pmod{3^P} \qquad (0 \leq j \leq 8).$$

If all nine A_j are certifiably excluded, every whole-block rotation of $12\cdots m$ fails to be a prime-exponent power over $\mathbb{Q}_3$, and consequently fails to be an integer perfect power. This is a sufficient test, not an asserted equivalence.

Proof. For a nonzero number of positive valuation b, a prime-exponent power can only have exponent dividing b. For exponent 2 its unit must be 1 modulo 3; for exponent 3 its unit must be ± 1 modulo 9. Thus the three conditions exclude every possible prime exponent. Since $T_{k+9} - T_k = 9k + 45$, the cut-dependent term of the cubic congruence in Theorem 7 depends only on $k \pmod 9$. That theorem identifies $10^{-D(k)}R_{m,k}$ with one of the nine residues to the required precision. Multiplication by $10^{D(k)} \equiv 1 \pmod 9$ preserves valuation and both unit tests. □

Corollary 5 (133 sufficient exclusion classes when $v_3(d) = 2$)**.** *Suppose $v_3(d) = 2$ and put $u = d/9$. Every rotation is excluded from prime-exponent powers over $\mathbb{Q}_3$ if*

$$m \bmod 9 \in \{2,3,5,6\}, \qquad \text{or} \qquad m \bmod 27 \in \{9,17\}, \qquad \text{or} \qquad m \bmod 243 \in E_u,$$

where

$$E_u = \begin{cases} \{26, 27, 81, 108, 134, 135, 216\}, & u \equiv 1 \pmod 3, \\ \{27, 81, 107, 108, 135, 215, 216\}, & u \equiv 2 \pmod 3. \end{cases}$$

The displayed sufficient set consists of exactly 133 classes modulo 243. No claim that it is the complete local exclusion set is made here.

Proof. Reduce the cubic congruence modulo 243. Since $d = 9u$, its residue is

$$F_u(m, k) = T_m + \frac{27u}{2}(m^3 - m + 9) - 81umT_k \pmod{243}.$$

It depends only on $m \pmod{243}$, $u \pmod 3$, and $k \pmod 9$: for the dependence on u, use $3 \mid (m^3 - m)$. The first two sets have valuation one, or valuation two with unit 2 modulo 3, just as in Corollary 4. For the additional residues $m = 27, 108, 135, 216 \pmod{243}$, respectively, the normalized unit $F_u(m, k)/27$ is $5, 2, 7, 4 \pmod 9$, independently of k and u. For $m = 81 \pmod{243}$ the valuation is four and the unit is 2 modulo 3. For $u \equiv 1 \pmod 3$, the residues $m = 26, 134 \pmod{243}$ have valuation three and respective normalized-unit sets $\{4, 7\}, \{2, 5\} \pmod 9$ as k varies. For $u \equiv 2 \pmod 3$, $m = 107, 215 \pmod{243}$ give the same two sets. These computations follow by substituting $T_k \pmod 3 \in \{0, 1\}$ in the displayed expression. None of the valuation-three units is ± 1 modulo 9. Valuations one through four have no prime divisor other than 2 or 3, so no further prime exponent is possible. The three displayed sets are disjoint and have sizes 108, 18, and 7, whose sum is 133. □

3 Lifting at boundary primes

Lemma 3 (Boundary periods)**.** *Let $q > 3$ be prime, $r = \operatorname{ord}_q(10)$, and $\gcd(r, 10) = 1$. Choose P with $r \mid P$ and $10^P \equiv 1 \pmod{9r}$. Then $S(d) \pmod r$ has period P, and $V_d \pmod q$ has period qP. More precisely, for $t \geq 0$ and $0 \leq s < P$,*

$$V_{tP+s} \equiv tV_P 10^{S(s)} + V_s \pmod q. \tag{15}$$

Proof. In a complete j-digit block let $a = 10^{j-1}$, $B = 10^j$, and $N = 9a$. Appending that block acts by $z \mapsto B^N z + U(j)$. Modulo q, if $B \neq 1$,

$$U(j) = \frac{(a(B-1)+1)B^N - (B(B-1)+1)}{(B-1)^2}; \tag{16}$$

if $B = 1$, it is $Na + N(N-1)/2$. The affine map therefore depends only on $a, B, N \pmod q$ and $N \pmod r$. These data repeat after P steps: $r \mid P$ fixes the first three, and $10^P \equiv 1 \pmod r$ fixes the last. Also $9j10^{j-1} \pmod r$ has period P, and

$$S(P) = \frac{(9P-1)10^P + 1}{9} \equiv 0 \pmod r$$

by the stronger modulus $9r$. Thus $S(d) \pmod r$ is periodic, and a complete P-cycle acts by $z \mapsto z + V_P$ modulo q. Appending s more blocks proves (15), and adding q cycles proves the asserted period of V_d. □

Proof of Theorem 4. We interpolate a complete P-cycle of digit blocks and average it over an arbitrary number of cycles, retaining division by that number. Put

$$\lambda = \frac{\log(10^r)}{r} \in p\mathbb{Z}_p, \qquad \omega = 10e^{-\lambda}, \qquad L = P\lambda.$$

Then $\omega^r = 1$, and its reduction has order r. For $s = 1, \ldots, P$ and $X \in \mathbb{Z}_p$, set

$$j = PX + s, \quad B = 10^s e^{LX}, \quad A = B/10, \quad N = 9A, \quad b = B - 1, \quad S_s(X) = \frac{(9j-1)B + 1}{9}.$$

Let $E_s = 9s10^{s-1}$ and $H_s = S(s)$ be fixed integers. Define

$$U_s(X) = \frac{(Ab+1)\omega^{E_s} e^{\lambda jN} - (Bb+1)}{b^2}, \qquad F(X) = \sum_{s=1}^{P} U_s(X)\omega^{-H_s} e^{-\lambda S_s(X)}.$$

For integral $X \geq 0$, the period conditions give $jN \equiv E_s \pmod r$ and $S(j) \equiv H_s \pmod r$. Thus $U_s(X)$ is the actual j-digit block and $F(X) = \sum_{j=PX+1}^{P(X+1)} U_j 10^{-S(j)}$. For $r \mid s$, remove the apparent singularity by

$$U_s(X) = AN + (Ab+1)\sum_{t\geq 0} \binom{N}{t+2} b^t. \tag{17}$$

Here $b \in p\mathbb{Z}_p$ throughout $\mathbb{Z}_p$; if $r \nmid s$, b is a unit.

Write $F(X) = \sum_{n\geq 0} c_n X^n$. The coefficient bound is

$$v_p(c_n) \geq n - v_p(n!) - \left\lfloor \frac{n+1}{p-1} \right\rfloor \geq \left\lceil \frac{p-3}{p-1} n \right\rceil \quad (n \geq 1). \tag{18}$$

To see this, each nth derivative of e^{LX} gains at least p^n. The same holds for the nth derivatives of λjN and $-\lambda S_s$, for $n \geq 1$: a derivative of the linear factor j loses at most one L, compensated by λ. Exponentiating preserves the property, as do inverses of unit denominators. In the tth summand of (17), an undifferentiated factor b contributes p, while a factor differentiated $i \geq 1$ times contributes p^i. Combining all factors gives at least $p^{\max(t,n)}$ for the nth derivative, before the denominator $(t+2)!$. It loses at most $\lfloor (t+1)/(p-1) \rfloor$ powers of p. For $t \leq n$ this gives the asserted bound, and for $t > n$ use that $t - \lfloor (t+1)/(p-1) \rfloor$ is nondecreasing. The tails converge with all derivatives. The normalizing exponential preserves the bound; division by $n!$ gives the first inequality. The second uses $v_p(n!) \leq (n-1)/(p-1)$.

For $M \geq 1$, the exact Stirling identity

$$\frac{1}{M}\sum_{i=0}^{M-1} i^n = \sum_{t=0}^{n} \left\{ {n \atop t} \right\} \frac{t!}{t+1} \binom{M-1}{t}$$

bounds its valuation below by $-\lfloor \log_p(n+1) \rfloor$. For $p \geq 7$, (18) exceeds this loss by at least one: if $k = \lfloor \log_p(n+1) \rfloor \geq 1$, then $n \geq p^k - 1$ makes its final bound at least $(p-3)k \geq k+1$; the case $k = 0$ is immediate. Termwise averaging therefore gives

$$\frac{1}{M}\sum_{i=0}^{M-1} F(i) \equiv c_0 \equiv C \pmod p.$$

The last equality follows by concatenating the first P blocks and using $r \mid S(P)$. Likewise $V_{PM} = 10^{S(PM)} \sum_{i=0}^{M-1} F(i)$ and $r \mid S(PM)$ give the exact valuation $v_p(M)$ and normalized unit $CM/p^{v_p(M)}$.

Finally put $d = PM$ and $\nu = v_p(10^r - 1) \geq 1$. Lifting the exponent gives $v_p(10^d - 1) = \nu + a$. In $S(d) = d10^d - (10^d - 1)/9$, the summands have unequal valuations a and $\nu + a$, so $v_p(S(d)) = a$. Since $r \mid S(d)$,

$$v_p(10^{S(d)} - 1) = \nu + a \geq a + 1.$$

The rotation identity (1) now proves both assertions in (5), including the factor contributed by the cut. □

Proof of Theorem 5. Before using $C \neq 0$, the preceding averaging argument gives

$$V_d \equiv C\frac{d}{P} \pmod{p^{a+1}}.$$

Put $B = 10^d$ and $t = B - 1 - m$. The removed t integer blocks all have length d, so the exact tail identity is

$$V_d = B^t W_m + \sum_{j=1}^{t} (B-j)B^{j-1}.$$

Lifting the exponent gives $B \equiv 1 \pmod{p^{a+1}}$. Thus the tail sum is $-t(t-1)/2$, and $t \equiv -m$ gives $W_m \equiv V_d + m(m+1)/2$ at this precision. Also $D(m) = S(d) - dt$ is divisible by both r and p^a; the latter follows from $S(d) = d10^d - (10^d - 1)/9$. Hence $10^{D(m)} \equiv 1 \pmod{p^{a+1}}$, and (1) proves (6).

When $p \mid d/P$, the polynomial in parentheses reduces modulo p to $m(m+1)/2$. Its roots 0 and -1 are simple. Each has p lifts modulo p^2, exactly one of which is zero modulo p^2. The other $p-1$ lifts have valuation one, proving the count. At $p = 37$, $P = 3$, $d = 111u$, the polynomial is $m(m+1)/2 + 1221u \pmod{1369}$; its two zero lifts are $296u$ and $1073u - 1$.

A further whole-block exclusion is $d \equiv 3 \pmod{111}$, $m \equiv 126 \pmod{999}$ within the d-digit interval. The elementary identities $W_m \equiv m(m+1)/2 + 9d\lfloor (m+1)/3 \rfloor \pmod{27}$ and $D(m) \equiv dm \pmod 9$ (proved in `RESIDUE_EXCLUSIONS.md`) give $R_{m,k} \equiv 9 \pmod{27}$, forcing the exponent to be 2. But (6) has polynomial residue 5 modulo 37, a nonsquare, and 10 is a square modulo 37. This excludes every rotation. □

Proof of Corollary 2. For a finite set $\mathcal{S}$ of primes $p \geq 7$ with $\gcd(\mathrm{ord}_p(10), 10) = 1$, choose $P_p = \mathrm{lcm}(r_p, \mathrm{ord}_{9r_p}(10))$. This satisfies the period hypotheses and $p \nmid P_p$: every prime factor of $r_p\varphi(9r_p)$ is smaller than p. Set

$$L = \mathrm{lcm}_{p\in\mathcal{S}}(P_p p), \quad M = \prod_{p\in\mathcal{S}} p^2, \quad \theta = \prod_{p\in\mathcal{S}} \left(1 - \frac{2(p-1)}{p^2}\right).$$

For fixed $L \mid d$, the Chinese remainder theorem shows that the fraction of classes modulo M avoiding every valuation-one obstruction is exactly θ. Thus, with $H = 9 \cdot 10^{d-1}$, at least $H(1-\theta) - M$ values of m have a valuation-one prime shared by every rotation, so none of their rotations is powerful.

Primes $p \equiv 3 \pmod{40}$ supply a divergent reciprocal sum by Dirichlet's theorem [4]. Quadratic reciprocity gives $(10/p) = 1$, and $(p-1)/2 \equiv 1 \pmod{10}$; therefore r_p is odd and coprime to 10. Consequently θ can be made smaller than $\varepsilon/2$. Choose such a finite set and then d_0 so that $M/H < \varepsilon/2$ for $d \geq d_0$. □

Proof of Theorem 3. For both pairs in Theorem 3,

$$\mathrm{ord}_p(10) = s \mid 9, \qquad v_p(10^s - 1) = 1, \qquad S(j) := D(10^j - 1) \in 9\mathbb{Z}.$$

Let U_j concatenate the j-digit integers, and put $B = 10^j$, $b = B - 1$, $a = B/10$, $q = 9a$. Its length jq is divisible by s. If $s \nmid j$, the block formula used in (11) gives $U_j \equiv -9a/b \pmod{p}$. If $s \mid j$, its multiplier B is one modulo p, so summing the appended integers gives $U_j \equiv 9/200 \pmod{p}$. Thus $U_j \pmod{p}$ has period s. Pairing each nonidentity sth root $z = 10^j$ with z^{-1} yields

$$\sum_{j=1}^{s-1} \frac{10^j}{10^j - 1} = \frac{s-1}{2}, \qquad \sum_{j=1}^{s} U_j \equiv -\frac{9(s-1)}{20} + \frac{9}{200} = C_s \pmod{p}.$$

All block multipliers are one modulo p, proving (4), including its rotation assertion. This proves Theorem 3. Its corresponding valuations and normalized units follow from Theorem 4 with $P = s$.

□

4 Power characters and periodic transfer

For a prime q and a prime $e \mid q - 1$, put $\chi_{q,e}(z) = z^{(q-1)/e} \pmod{q}$. A nonzero eth power has character 1. If

$$10^{S(d)} \equiv 1 \pmod{q}, \qquad \chi_{q,e}(10) = 1, \qquad \chi_{q,e}(V_d) \notin \{0, 1\}, \tag{19}$$

then (1) gives the same nontrivial character for every $R_{10^d-1,k}$. Thus none is an eth power. When $v_3(d) + 2$ is a power of e, Theorem 2 eliminates all other prime exponents.

Here are the finite inputs. Their nonzero cycle constants also verify every hypothesis of Theorem 4:

i	q_i	r_i	P_i	$V_{P_i} \pmod{q_i}$
1	37	3	3	33
2	757	27	27	538
3	333667	9	9	221885
4	440334654777631	27	27	178343635184937
5	1527791	43	903	547718
6	43037	29	812	409

For each row of the next table, $r_i \mid S(d)$, $\chi_{q_i,e}(10) = 1$, and the displayed character is nonzero and different from 1. Each $v_3(d) + 2$ is a power of the listed e.

d	e	i	$\chi_{q_i,e}(V_d)$
2, 14, 20	2	2	−1
5, 26	2	3	−1
8, 11, 17, 18	2	1	−1
12, 15	3	1	26
23	2	4	−1
27	5	5	990998
54	5	4	386179030766766
243	7	6	12819

The accompanying `BOUNDARY_CHARACTERS.json` records these exact inputs; `verify_boundary_characters.py` verifies them. Their finite arithmetic consists of complete factorizations of $q_i - 1$, Lucas primality witnesses, modular powers, and the recurrence $V_j = B^N V_{j-1} + U(j)$ with (16). An independent verification uses exact matrix powering for every listed d. For a Lucas witness a, the checked conditions are $a^{q-1} \equiv 1 \pmod q$ and $\gcd(a^{(q-1)/\ell} - 1, q) = 1$ for every prime $\ell \mid q-1$; the factor primes are verified by trial division. These conditions force every prime divisor of q to be at least q, proving primality.

Proof of Theorem 6. Corollary 1 and the table exhaust $d = 1, \dots, 27$; the last two rows handle 54 and 243. Equation (19) excludes every available prime exponent.

For any row put $h = v_3(d)$ and $T = q_i \operatorname{lcm}(P_i, 3^{h+1})$. Replacing d by $d + tT$ preserves h, $S(d) \pmod{r_i}$, and $V_d \pmod{q_i}$, by Lemma 3. Hence the same character argument works for all $t \geq 0$. The rows $d = 12, 15, 27, 243$ give exactly (7). □

Corollary 6 (Exact local power criteria)**.** *Put $Q = 440334654777631$ and let u be a positive integer. For every rotation at the indicated boundary,*

d	*equivalence over the stated field*
$27u$	R *is a fifth power in* $\mathbb{Q}_Q \iff u$ *is a fifth power in* $\mathbb{Q}_Q$
$243u$	R *is a seventh power in* $\mathbb{Q}_{757} \iff u$ *is a seventh power in* $\mathbb{Q}_{757}$.

Explicitly, for $(p, e) = (Q, 5)$ or $(757, 7)$, write $u = p^a t$ with $p \nmid t$. The local condition is exactly

$$e \mid a, \qquad t^{(p-1)/e} \equiv 1 \pmod p. \tag{20}$$

If $3 \nmid u$, failure of this condition excludes every integer perfect power in the whole rotation block.

Proof. Theorem 4 gives $v_p(R) = v_p(u) = a$ and

$$R/p^a \equiv 10^{D(k)} 178343635184937\, t \pmod Q, \qquad R/p^a \equiv 10^{D(k)} 300t \pmod{757},$$

respectively. The displayed coefficients and 10 have character 1 for exponents 5 and 7 respectively, as checked in the certificates. Since $e \neq p$, Hensel's lemma lifts every nonzero eth root modulo p. Thus a number in $\mathbb{Q}_p$ is an eth power exactly when its valuation is divisible by e and its normalized unit is an eth power modulo p. This proves both equivalences. When $3 \nmid u$, the only possible prime exponent is 5 or 7 respectively, by Theorem 2. □

Proof of Corollary 3. The 3-adic restrictions leave only the following disjoint sets:

$$d \equiv 2 \pmod 3, \qquad d \equiv 12, 15 \pmod{27}, \qquad d \equiv 18 \pmod{27}, \qquad 27 \mid d.$$

In the first and third sets, a perfect power must be a square; in the second, a cube. Put $Q = 440334654777631$. In every fixed class modulo 27, Lemma 3 makes V_d affine modulo each of $37, 333667, 757, Q$. In the class $d \equiv a \pmod{27}$ the respective slopes are $1, 331988, 538, 178343635184937$ multiplied by $10^{S(a)}$; all are nonzero. The four residues are independent by the Chinese remainder theorem. For a square candidate the exact valuation of $S(d)$ from (12) shows $27 \mid S(d)$. Thus all four rotation characters are fixed, and the proportion of residues that are zero or squares at all four primes is

$$\alpha = \frac{19}{37}\frac{166834}{333667}\frac{379}{757}\frac{Q+1}{2Q}.$$

For cubes use 37 and 333667, giving the proportion $\beta = (13/37)(111223/333667)$, as in (4). The upper density of the remaining d is therefore at most $(10\alpha + 2\beta + 1)/27$.

For a prime $e \mid p - 1$, the density of positive integers that are eth powers in $\mathbb{Q}_p$ is

$$s(p, e) = \sum_{j \geq 0} \frac{p-1}{e p^{ej+1}} = \frac{(p-1)p^{e-1}}{e(p^e - 1)}.$$

Indeed, the valuation must be a multiple of e, and exactly $(p-1)/e$ normalized units modulo p are eth powers. Truncating at valuation K leaves a subset of multiples of p^K, of density p^{-K}, proving the density assertion. For $p \neq 3$, the same proportion applies conditional on $3 \nmid u$, by the Chinese remainder theorem and the same truncation. Thus, inside $27 \mid d$, Corollary 6 excludes disjoint sets of densities

$$\delta_5 = \frac{2}{81}\big(1 - s(Q, 5)\big), \qquad \delta_7 = \frac{2}{729}\big(1 - s(757, 7)\big).$$

Here $v_3(d)$ is respectively 3 or 5; positive local valuations are included. Also $d = 81u$ with $u \equiv 1, 7 \pmod 9$ is excluded by Corollary 1, adding the disjoint density $2/729$. For $h = v_3(d) \in \{6, 7, 10, 14, 16\}$, Corollary 1 adds disjoint exclusions of total density

$$E = \frac{1}{2187} + \frac{4}{19683} + \frac{2}{531441} + \frac{1}{14348907} + \frac{2}{387420489}.$$

These follow respectively from $h + 2 = 8, 9, 12, 16, 18$ and the indicated unit classes in that corollary. None overlaps the already counted sets, which have $h \le 5$. Consequently the exact rational lower bound is

$$\rho_* := 1 - \frac{10\alpha + 2\beta + 1}{27} + \delta_5 + \delta_7 + \frac{2}{729} + E > 0.95599527150.$$

The companion `verify_density_refinement.py` checks this inequality using rational arithmetic. □

5 A uniform sieve for powerful values

We now prove Theorem 1. In a fixed d-digit interval set

$$B = g^d, \quad L = g^{d-1}, \quad c = B - 1, \quad A = c^2 W_{L-1} + Lc + 1.$$

The recurrence $W_{m+1} = BW_m + (m + 1)$ gives, for $L - 1 \le m \le B - 1$,

$$c^2 W_m = AB^{m-L+1} - ((m + 1)c + 1), \qquad D(m) = D(L - 1) + d(m - L + 1).$$

For fixed k, writing $u = g^{D(k)}$, substitution into (1) gives

$$R_{m,k} = \alpha B^m + \beta m + \gamma, \quad \begin{cases} \alpha = (uA/c^2 - g^{D(L-1)} W_k) B^{1-L}, \\ \beta = -u/c, \quad \gamma = W_k - uB/c^2. \end{cases} \tag{21}$$

At every prime $p \nmid g(B - 1)$ the coefficients are defined modulo p^2 and β is a unit. The right side provides a periodic modular extension; only its values inside the digit interval are identified with the concatenation.

Lemma 4 (Valuation-one densities)**.** *Let $p \nmid B\beta$ and $F(n) = \alpha B^n + \beta n + \gamma \pmod{p^2}$. The event $p \mid F(n)$, $p^2 \nmid F(n)$ has period dividing $T_p = p^2(p - 1)$ and density $\rho_p = (p - 1)/p^2$. For any finite set P of eligible primes the events have joint density $\prod_{p \in P} \rho_p$ in a period dividing $\operatorname{lcm}_{p \in P} T_p$.*

Proof. Fix $n \pmod{p - 1}$. The exponential term is constant modulo p, and the linear coefficient is nonzero, so exactly one class $n \pmod p$ is a root. Its p lifts differ by $p(p - 1)t$. Euler's theorem fixes the exponential term modulo p^2 on these lifts, while the linear term takes p distinct values. Precisely one lift is zero modulo p^2, leaving $p - 1$ of valuation one.

For the joint assertion induct on the largest prime q. The smaller-prime events have a period T with no prime divisor as large as q. Thus $q \nmid H = \operatorname{lcm}(T, q - 1)$. In each fixed class modulo H, the smaller-prime events are fixed and the Chinese remainder theorem lets $n \pmod{q^2}$ vary freely. The local count gives the same fraction ρ_q in every class. Averaging multiplies the previous joint density by ρ_q. This allows overlapping periods, including a smaller prime dividing a larger prime minus one. □

For an interval I of H integers and a subset P', Lemma 4 gives its simultaneous event count as $H \prod_{p \in P'} \rho_p + O(\prod_{p \in P'} p^3)$, since the interval error is at most the joint period. Take eligible primes at most z, put $\lambda = \sum \rho_p$, and choose an even integer $K \ge 0$. Powerful values avoid every valuation-one event. Finite Bonferroni gives

$$\#\{n \in I : F(n) \text{ powerful}\} \le H \left(e^{-\lambda} + \frac{\lambda^{K+1}}{(K + 1)!} \right) + (K + 1) z^{4K}. \tag{22}$$

Indeed, if e_j are the elementary symmetric sums of the ρ_p, the even truncated main term is at most $\prod(1 - \rho_p) + e_{K+1} \le e^{-\lambda} + \lambda^{K+1}/(K + 1)!$. There are at most z^j subsets of size j, each costing at most z^{3j}, which proves the displayed error. These estimates are uniform in the coefficients of (21) and in k.

Choose

$$\eta = \frac{\log g}{100}, \qquad z = \exp\left(\frac{\eta d}{\log d} \right), \qquad K = 2\lceil 4 \log d \rceil,$$

and use primes $d^2 < p \le z$ with $p \nmid g(g^d-1)$. For large d, $z > d^2$. Omitted primes exceeding d^2 and dividing g^d-1 number at most $\log(g^d-1)/(2\log d)$, with reciprocal sum $O_g(1/(d\log d))$. The fixed prime divisors of g eventually lie below d^2. Mertens' estimate [3] therefore yields

$$\lambda = \log\log z - \log\log(d^2) + O_g(1) = \log d - 2\log\log d + O_g(1).$$

In particular $0 < \lambda \le \log d$ eventually, so $e^{-\lambda} \ll_g (\log d)^2/d$. Also

$$\frac{\lambda^{K+1}}{(K+1)!} \le (e/8)^{K+1} \le d^{-8}, \qquad (K+1)z^{4K} < g^{d/2}$$

for large d; the latter follows from $K \le 10\log d$ and $\log((K+1)z^{4K}) \le \log(K+1) + 40\eta d < d\log g/2$. Thus for every fixed k and every interval inside the digit block with $m > k$,

$$\#\{m \in I : R_{m,k} \text{ powerful}\} \ll_g |I|\frac{(\log d)^2}{d} + g^{d/2}. \tag{23}$$

Let D be the digit length of $\lfloor X \rfloor$. For each $d \le D$ and k, apply (23) on $\max(g^{d-1}, k+1, 2) \le m \le \min(X, g^d-1)$. There are at most g^d choices of k, so the total interval error is $\sum_{d\le D} g^{3d/2} = O_g(X^{3/2})$. Blocks with $d \le D/2$ contain only $O_g(X)$ pairs and may be bounded trivially. On the other blocks the main factor is $O((\log D)^2/D)$, and there are $O(X^2)$ pairs. This proves (2).

Finally, each m supplies m distinct rotations. If m has $d \ge 2$ digits, the cyclic digit word has a unique occurrence of 1 followed by $d-1$ zeros: it is the integer g^{d-1}. Such a zero run cannot cross an integer boundary or the cyclic join, whose next digit is 1. The unique marker prevents a nonidentity digit rotation from fixing the word. For $d = 1$ all digits are distinct. Digit lengths increase strictly with m, so the sequence through block $m = X$ has $X(X+1)/2$ entries. Bounding a partial final block by the whole block and taking $X \asymp \sqrt{N}$ proves the sequence-index assertion. □

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